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Increasing Reliability and Lowering Maintenance of Offshore Surface Wellhead Control Systems Using All-Electric Controls

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Abstract

There is a current push for electrification in industrial applications. First came manual operations, then steam powered, next advanced pneumatic and hydraulic control systems and now electrification. As the cost of motors has come way down, electric actuators can be a more viable solution due to their long-life span and low maintenance. But electric actuators are not only preferred for their lower maintenance, their operation when needed philosophy and simple control systems also require less power demand from the site power distribution. This means that power sizing can be significantly reduced, and power consumption will be less. Each one of these features contributes to a lower overall ESG score for offshore assets. When replacing hydraulic control systems with their electric counterparts there are some design advantages that enable equipment providers to increase systems reliability. The transition from hydraulic to electric actuators in industrial applications not only enhances system reliability and reduces maintenance, but also significantly lowers power consumption and ESG impact, making electrification a strategic and sustainable advancement for offshore asset design.

This paper discusses the design and delivery of electric actuators and their corresponding control system on a surface drilled oil and gas drilling and production platform. Based on customer requirements, a highly reliable electric actuator control system was designed and delivered. The system enables normally unattended operations by reducing maintenance needs for offshore surface drilling activities while also decreasing power requirements. Lower maintenance and power consumption are inherent features to the electric actuator and control system, but further design choices can be made to achieve significant reliability improvements in a tradition wellhead control system. A robust and carefully thought-out power distribution was utilized to align with the customer requirements. Redundancy PLCs as well as networking components were used to form a redundant network. Lastly, small form factor motor servo drives were connected in a redundant networking architecture to ensure that no single fault in the servo drives could disrupt the operations of the system. The result produced a small footprint but highly redundant control system which saved platform space and lowered installation costs compared to legacy hydraulic control system. The implementation was successful, significantly enhancing system reliability and reducing maintenance needs. Furthermore, this system and the respective installation is the first of its kind in the world where an electric actuator has been utilized on an offshore surface drilled wellhead and christmas tree. The design,

development and ultimate installation proved the simplicity and reliability of electric actuators and their respective control systems versus their hydraulic predecessors.

Introduction

Traditional hydraulic wellhead control systems (WHCP) have reached their technical peak. Hydraulic actuators operated with the latest designed WHCP are electric over hydraulic utilizing smart PLC based control systems and field mounted instrumentation to give operators remote operation and feedback on the system. This design enables unmanned operation, but it doesn't remove the headache of the hydraulic assemblies and control instrumentation and the high maintenance requirements that they demand. Therefore, wellhead maintenance technicians must still travel out to the platform to perform interval inspections and maintenance to keep the system running. Hydraulic control systems are also inherently not fault tolerant. To design as efficiently as possible the power source, hydraulic control fluid, is contained and pressurized at one point and then distributed out to each function. Any failure in this single distribution causes a system wide failure. By replacing hydraulic actuators with electric, the control system can be all-electric. This replaces the high demand maintenance parts with non moving and highly reliable electric controls. This enables a true unmanned operation. Providing remote operation and monitoring while removing maintenance needs. The next step is to improve system reliability. The system we designed makes the power distribution and communication single fault tolerant at a minimum. Thus any one fault will have no effect on the system or status of the platform production.

Statement of Theory and Definitions

The feedback that operators can currently get from hydraulic actuators are valve position status and hydraulic system pressure. A failure in either one of these could be caused by any number of system level failures. Investigating the hydraulic system pressure for example, the hydraulic system pressure could drop below the required setpoint to keep the valve in the Open position. This could be caused by a leak in the connection at the actuator, a leak at the solenoid supplying the actuator, a leak in the manifold supplying the solenoid or a leak in the pipework between the manifold and the pump output. To diagnose and troubleshoot these types of issues takes time. Some of these failures could even impact the entire system as each actuator is being sourced from a manifold or head pressure. If this one pressure was to drop all actuators could begin to close from spring force. Ultimately production flows would be affected. Looking at the open and close position status this does in fact tell you about the individual actuator, but it does not provide insight into the reason why the actuator may have shifted into a non-desired position.

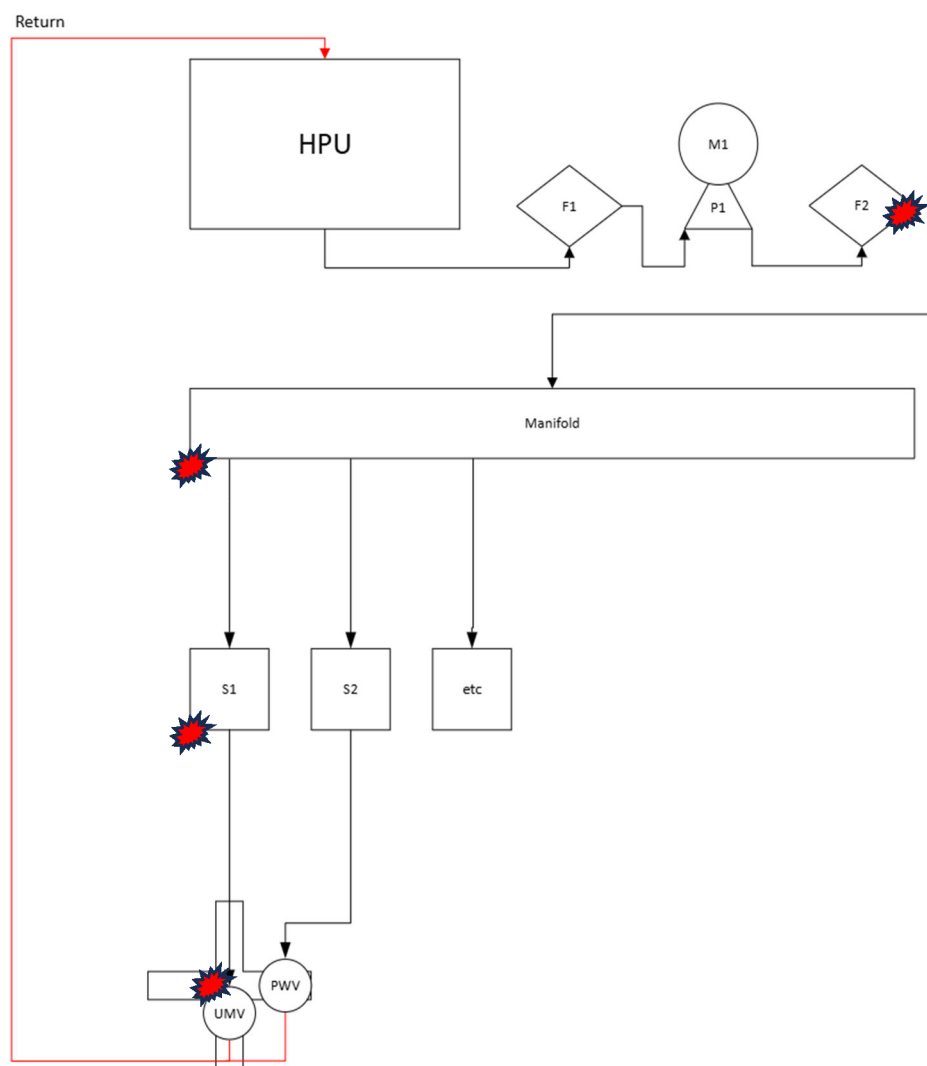


Figure 1—Single Faults which could affect actuator position.

One of the key advantages electric actuators have are discrete monitoring capabilities. This means that many individual features of each valve and actuator can be monitored for a lower level of condition monitoring. Think system level for hydraulic actuators but component level for electric actuators. Electric actuators are outfitted with numerous sensors telling the operator the health of the exact valve and actuator. These sensors are directly connected to the control system and reported. If any individual failure occurs, or the sensor begins to show operation outside of the normally configured operating range, the operator is reported of the exact failure mode. All troubleshooting time is saved by knowing exactly where the issue exists. Each failure is relevant to one valve at a time and does not affect other valves in the system. This allows the operator to plan for maintenance and ensure they are taking care of the right part of the system. Furthermore, traditional fieldbus Protocols can be used to tag all pertinent information so that it is readily available for features such as online dashboards which are utilized by wellhead maintenance teams. This data can be available on the platform but also online where it could be evaluated by cognitive AI that will learn to spot trends over time and offer advice on what maintenance is required.

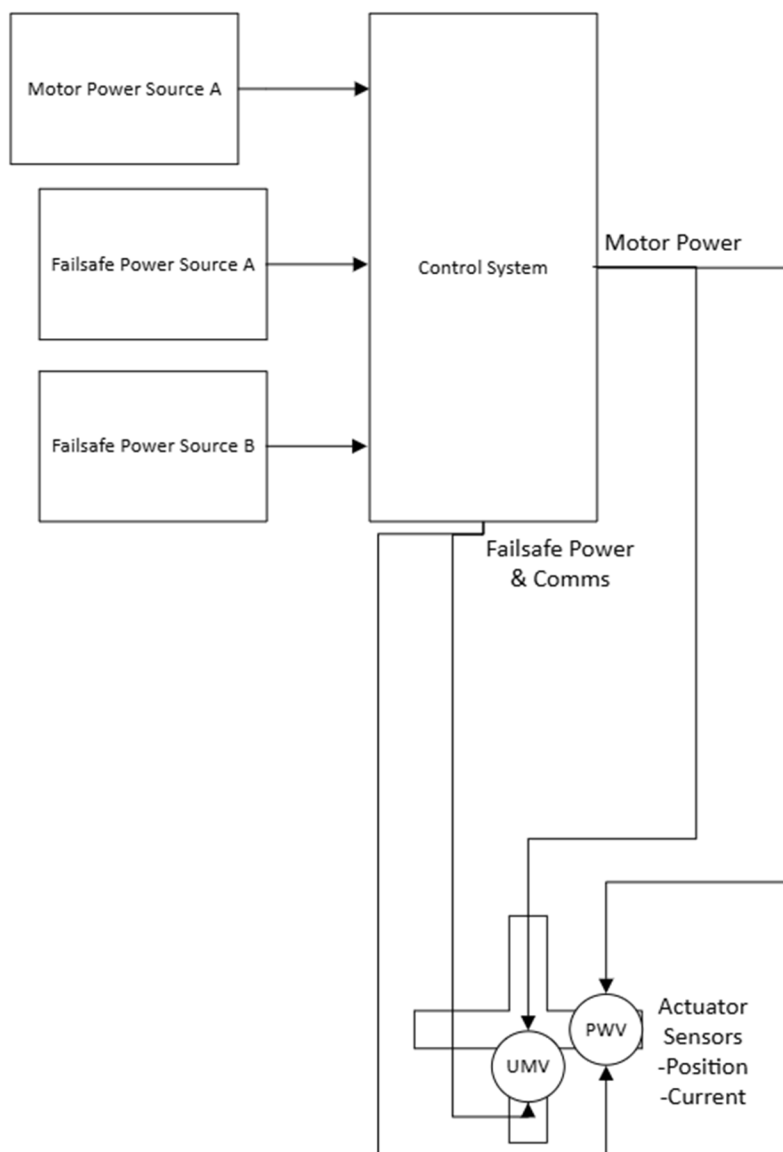


Figure 2—Actuator Power and Communication System Level

The usage of PLC networking components and motor servo drives in redundant architecture for the electric actuator control system adds fault tolerance to the system. In the final design, even if one of the servo drives was to fail the system re-routes all traffic to the other servo drives so that only one actuator can be affected at a time.

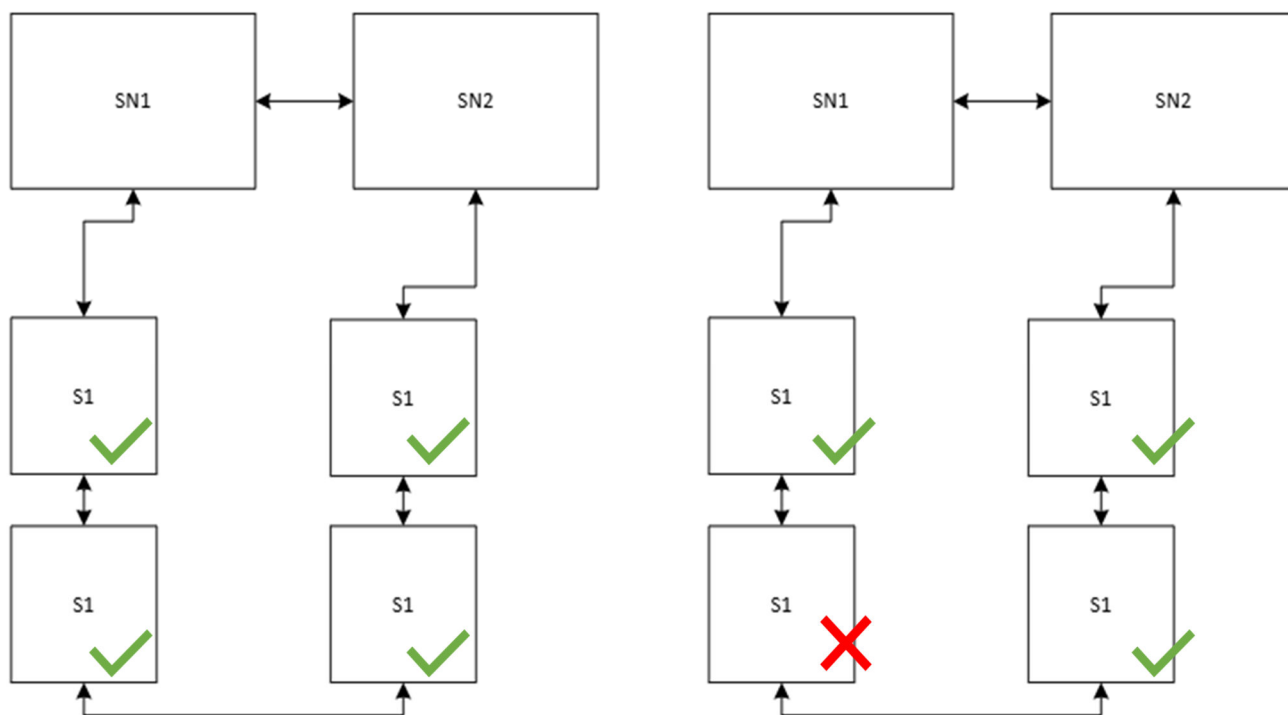
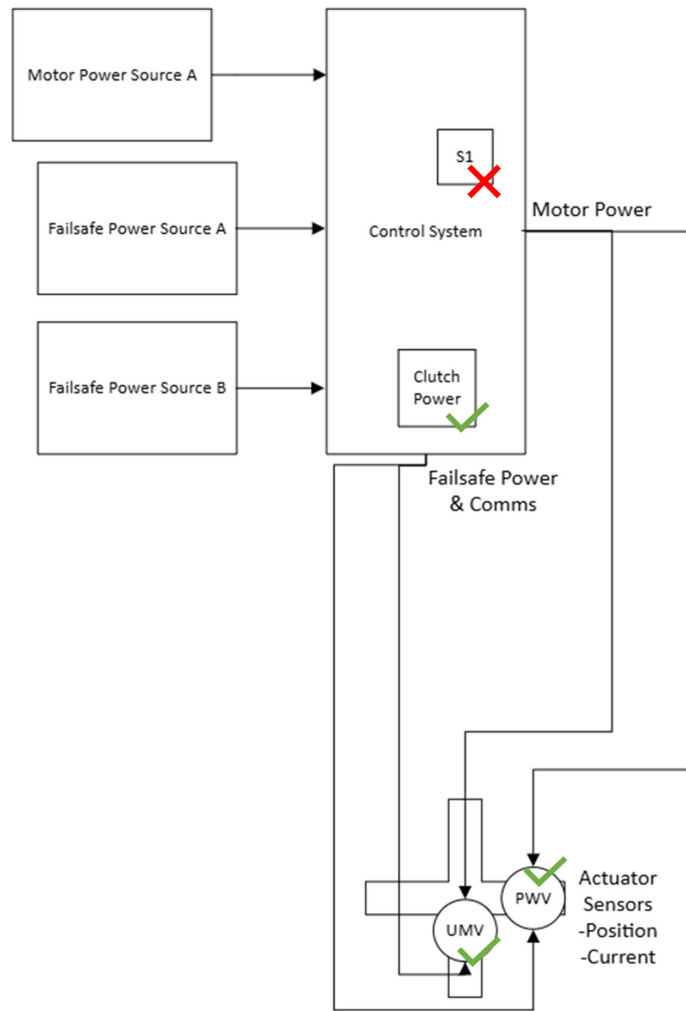


Figure 3—Failure of one Servo Drive does not affect any other actuator

Additionally, the actuator itself is still fail-safe in design. Meaning the servo drive is only responsible for motor open and close operation. The actuator has a secondary means to function, which is completely decoupled from the servo motor control and actuator motor drive train. This is a fail-close spring method. The drivetrain remains engaged by the usage of a clutch and this clutch is directly powered from the customers' ESD system on the platform. Even if the servo drive was to fail and the operator losses normal control of the actuator motor the ESD system could still remove power from the clutch and effectively de-couple the drive train from the valve operating gear which "fails close" the valve in the event it was needed. If the scenario did not present itself then the actuator would remain open, held in place by the clutch, and there would be no impact on the production flow. When the servo drive is lost that one affected actuator essentially fails to a state that it operates similar to a hydraulic actuator does today, held open until the open force is removed.



Hydraulic control systems require regular and extensive maintenance to ensure optimal performance and safety. Key maintenance tasks include:

- **Oil Maintenance:** Regularly checking and replacing hydraulic oil to keep it clean and free of contaminants
- **Filter Changes:** Frequently changing filters to prevent clogs and maintain system efficiency
- **Seal Replacement:** Periodically replacing seals to prevent leaks and maintain pressure
- **Inspection of Components:** Regularly inspecting hoses, fittings, and other components for wear and tear
- **Preventive Maintenance:** Implementing a preventive maintenance program to identify and address potential issues before they lead to system failures

Hydraulic systems are prone to leaks, which can cause significant downtime and require immediate attention. The high pressures and temperatures associated with hydraulics also necessitate careful monitoring and maintenance to prevent safety hazards while working in the red zone.

Electric control systems, on the other hand, have lower maintenance requirements compared to hydraulic systems. Key maintenance tasks include:

- **Visual Inspections:** Regularly inspecting electrical connections, wiring, and components for signs of wear or damage

- **System Testing:** Periodically testing the system to ensure all components are functioning correctly
- **Cleaning and Lubrication:** Keeping components clean and lubricated to prevent overheating and ensure smooth operation
- **Monitoring Power Distribution:** Ensuring consistent power supply and addressing any issues with power distribution

Electric systems benefit from fewer moving parts and the absence of fluid leaks, resulting in reduced maintenance needs and increased reliability. The use of redundant networking architectures further enhances fault tolerance and minimizes the risk of system failures

Many of the activities listed above, such as system testing and monitoring of power distribution, can be performed remotely with this electric control system. For the visual inspection and cleaning and lubrication extensive cycle testing was performed to ensure the actuator was designed with a field life more than 30 years. The monitoring of specific actuator parameters ensures that the actuator is operating the same as it did during day 1 commissioning.

Maintenance Event	Frequency					Time Required	
	3 M	6 M	1 YR	2 YR	5YR	Per Visit	Annual
Inspection of leak points	✓					4 hrs	16 hrs
Oil Sampling	✓						
Flush/Replace Filters and Strainers		✓				4 hrs	8 hrs
Grease low voltage motors		✓					
Pump Inspections at HPU			✓			52.5 hrs	52.5 hrs
Reservoir Inspection and maintenance			✓				
Accumulator inspection and maintenance			✓				
Space heater inspection			✓				
Low voltage motor inspection			✓				
Level transmitter inspection			✓				
Pressure transmitter calibrations			✓				
Temperature Switch calibrations			✓				
Function test Wellhead Control Panel				✓		1.5 hrs	.75 hrs
Function test all Pushbuttons				✓			
Replace all hoses				✓		Capital Costs	
Recertify all accumulators					✓		
Annual Subtotal							77.25 hrs
Annual Total (4-person crew)							309 hrs

Figure 4—Annual maintenance of large scale hydraulic control system per operator interviews

Methods Procedures, Process

The project involved using fail safe electric actuators for remote-controlled gate valves on surface wellheads. These actuators were designed to be able to close and shear wireline in the event of an emergency. A motor control system panel was designed to operate the new electric actuators, with one panel capable of operating eight actuators within a small form factor footprint. The panel was also integrated with the platform SCADA system, enabling local and remote control and monitoring of actuator and valve health.

The panel received commands via Profibus DP and shared status via OPC-UA. It also provided SIL3 rated interfaces for Emergency System Disconnect (ESD) or Platform Safety Shutdown (PSD) events. These were designed with unique shutdowns either per well or per actuator as needed. Custom circuits were designed which allowed a local handheld shutdown, remote shutdown and a local and remote reset after shutdown. The remote reset was particularly important for normally unattended operations. If a failure scenario was to occur, and the event was handled or cleared, the operator needed the ability to open the valve backup and resume production safely without having to send people out to the platform.

The panel accepted local 400VAC from the platform and distributed this power to two redundant 24VDC power supplies for internal PLC and instrumentation. Additionally, it accepted another completely redundant 24VDC power from a UPS on the platform, ensuring a multi-redundant PLC power supply and 24VDC bus within the panel. Each of the three power supplies were diode or'd together and the system sized appropriately to maintain backplane power and clutch control in the event one power supply failed.

The 400VAC was used for actuator open and controlled close operations while the 24VDC was used to power all PLC and networking components as well as provide power for the fail close mechanism on the actuator, which is the clutch. In the event the main 400VAC was lost from the panel all communication remains healthy and active, the actuators remain open (meaning no disruption to production) and the ability to close the actuator if needed remains. The platform UPS is sized to maintain this status for enough time to restore the 400VAC power. The energy capacity demand is kept low due to the low power demand of the actuator.

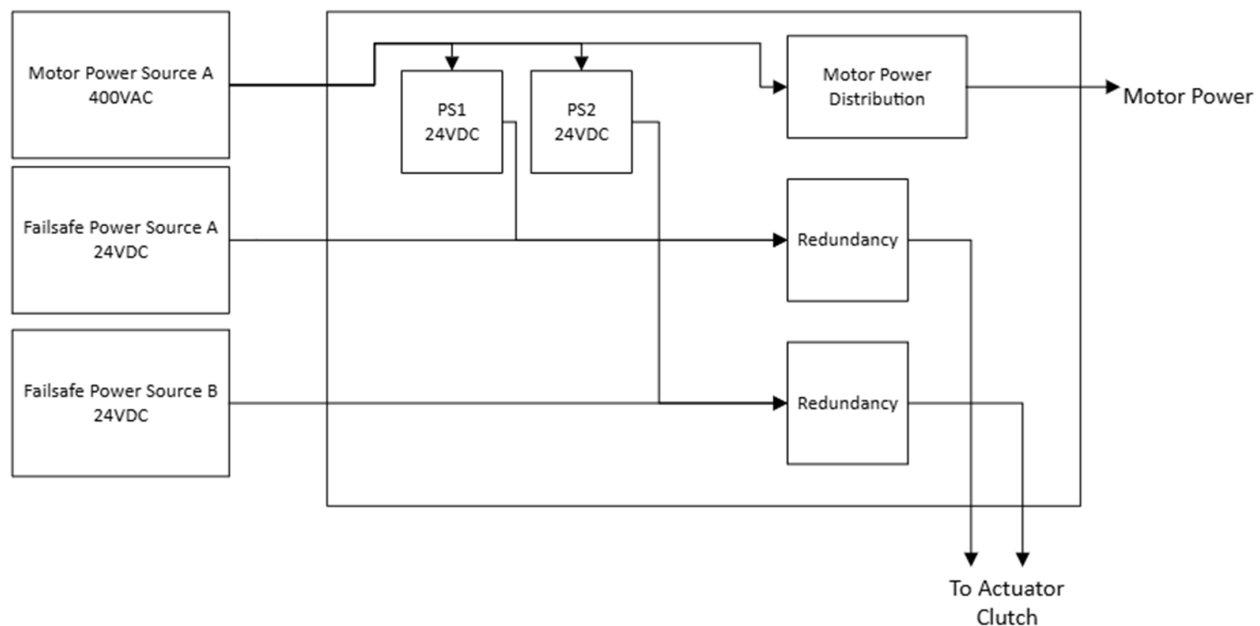


Figure 5—Clutch Power Redundancy

The motor control system also featured a novel redundant PLC network, including two PLCs, a PN/PN Coupler, and two redundant I/O racks. The system was designed with one "hot" rack and the other on "warm standby" such that it could quickly failover from one rack to the other in the event of an IO card failure,

ensuring continuous operation and data sharing. This was a pertinent feature for the desired not normally attended operation.

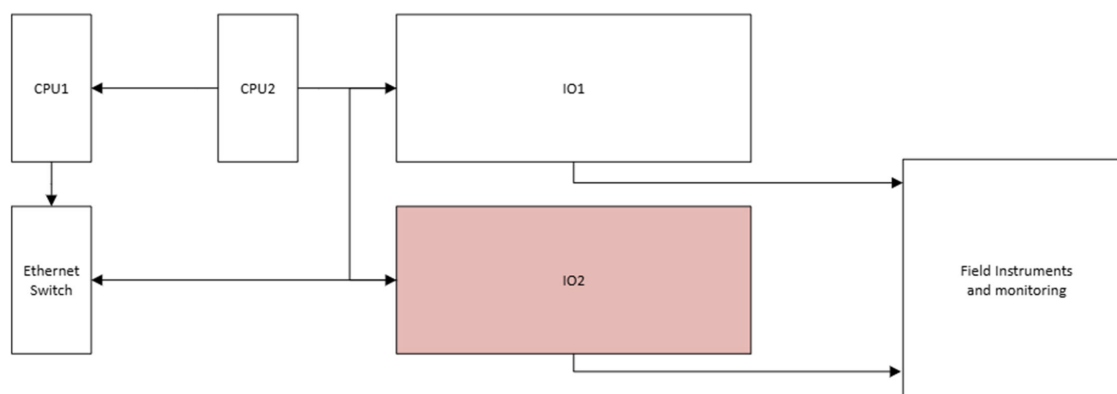


Figure 6—Redundant IO Network with one on warm standby

The servo drives for the motors were set up in a redundant Profinet EtherCAT network. This network was designed using two Profinet EtherCAT Gateways (Primary and Secondary), two Gateway Controllers (Primary and Secondary), and two EtherCAT switches. The communication between the PLC and servo drive continued to operate even if the EtherCAT ring network experienced a break in communication or if a single Profinet gateway failed. Lastly, as described above, if one individual servo drive was to fail all other servo drives would remain online.

Additionally, a local Human-Machine Interface (HMI) was designed and supplied for onsite commissioning and troubleshooting. This HMI facilitated easy monitoring and control of the system, ensuring efficient and effective operations.

Lastly, a bespoke GUI was created which offered many unique features such as alarm monitoring, trending local commissioning and even routines such as partial stroke testing and well testing scenarios.

All alarms are logged on an alarms page and each screen have a flashing alarm banner which gives insight of the occurrence. When an alarm is acknowledged the flashing will go away but the alarm will stay on the log until it is formally cleared from the system. Each valve cycle is trended, and a file is saved. Historical files can be accessed and reviewed as needed. At the time the valves and actuator are commissioned a valve signature can be generated. Thus, subsequent tests of the actuators can be compared against the signature to notify operators of any anomalies. With such a long actuator lifespan this is a great feature to have as different crews may review the valve status over years, and the valve signature details may get lost over time. The local commissioning mode allows an engineering user to function each individual component of the actuator if any configuration is needed over time. This is useful if adjustments are needed. Partial stroke testing can be configured to close the valve to a certain small percentage, which allows the entire actuator drive train to be energized and used, and all discrete features of the active components will be checked. Once the desired close percentage is reached the valve will function back to the open position. One day this type of test may be able to prolong infield valve testing since the operator can reliably test that the actuator is working and shall close as needed.

But when valve tests are required it is good to know that specific test scenarios can be programmed into the control system. This eliminates human error and provides a safe and reliable way to perform well tests, even remotely.

For example, for a christmas tree startup it can be programmed to first close the wing valve, then open the upper master valve, then open the lower master valve, then get status that the downhole safety valve is open prior to opening the wing valve backup and allowing the production to flow through the actuated choke.

Results, Observations, Conclusions

The electric actuator control system demonstrated significant improvements in reliability and fault tolerance compared to traditional hydraulic control systems. The redundant network architecture ensured continuous operation even in the event of a single component failure. The system's discrete monitoring capabilities provided detailed insights into actuator and valve performance, enhancing safety and maintenance efficiency.

The implementation of an all-electric actuator control system for offshore surface wellheads has proven to be highly successful. The systems has significantly increased reliability and reduced maintenance requirements.

This system was installed on a working production platform and has been in operation for over 12 months without issues to date. The platform is developed with a focus on sustainability and operational efficiency. It is a novel platform with power coming from a nearby windfarm. This integration of renewable energy reduces carbon emissions by approximately 85% over the project's lifecycle.

The platform is designed as a Not Normally Manned Installation (NNMI) and an Autonomous Production Location (APL), emphasizing minimal human intervention and high automation.

This platform's electrification aligns seamlessly with the electric actuator control system. The reliance on wind energy and its autonomous design necessitates highly reliable, low-maintenance systems with a low peak power requirement. The electric actuators, with their discrete monitoring capabilities and reduced need for manual intervention and very low peak power requirement are ideally suited for such an environment. The integration of redundant PLC networks and servo drives ensures that the system remains operational even in the event of component failures, which is critical for a platform designed to operate without regular human presence.

From a technical perspective, the platform's power system is designed to support the high demands of offshore gas production while maintaining sustainability. The use of wind energy eliminates the need for diesel generators, reducing operational costs and environmental risks. The platform's control systems are integrated with advanced SCADA systems, enabling remote monitoring and control, which is essential for its NNMI status.

The electric actuator control system was developed with similar principles in mind. Its ability to operate reliably in remote, not normally attended environments makes it an excellent fit for platforms like this. The system's modular design, small footprint, and robust fault tolerance ensure that it can be deployed in space-constrained offshore environments without compromising performance.

When compared to a traditional hydraulic control system the electric system can provide benefits as shown in the table below.

	 Energy reduction	 Peak power sizing reduction	 CO ₂ e savings
HP/MP/LP Motors	< 0.1	24.3	0.1
Lighting/Heating	55.4	6.3	39.1
DC Bus	26.3	3.0	18.6
Actuator Motor	< 0.1	1.4	< 0.01
Actuator Clutch	9.5	1.1	8.9
Control Panel	7.9	0.9	5.6
Difference	64 MWh	30 kW	45 tons/year

Figure 7—Hydraulic vs Electric Power Consumption and Power Sizing Reductions

This platform exemplifies the future of offshore gas production as it is sustainable, automated, and highly efficient. The integration of the electric actuator control systems into such platforms not only enhances operational reliability but also supports broader environmental and energy transition goals. This case underscores the importance of innovative control technologies in advancing offshore energy infrastructure.